Advanced Fixed Income and Credit

Group Assignment

Part I: Deutsche Bank — Finding Relative-Value Trades (HBS 205-059)
Part II: Python Labs — US Treasury Yield Curve Reaction to Trump Tariff Threats

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1 Part I — The Deutsche Bank Case Study

Setup and conventions

Settlement date: **August 15, 2003**. Exhibit 1 lists 52 Treasury coupon bonds whose maturities fall on successive semi-annual dates (Feb 15 / Aug 15) from 2004 to 2029 with no gaps, i.e. exactly one bond at each of the nodes $t = 0.5, 1.0, \ldots, 26.0$ years. This allows the classic *successive substitution* bootstrap: because each bond introduces exactly one new unknown discount factor, we solve sequentially from the shortest to the longest maturity.

Compounding convention. Exhibit 4's footnote states explicitly that Deutsche Bank's own model yields are *Bond-Equivalent Yields* (BEY), i.e. twice the semi-annual compounded yield. Since U.S. Treasuries pay semi-annual coupons, we derive our own zero-coupon yields in the identical BEY convention throughout, so that every comparison against the DB model in Q1.2 is apples-to-apples (a continuously-compounded convention would have introduced a small but systematic bias in one direction).

$$\text{BEY} = 2\left(D^{-1/k} - 1\right), \qquad D = \text{discount factor to period } k \text{ (6-month steps)}.$$

Q1.1 — Bootstrapping the discount curve

Methodology. For bond k (maturing at period k, coupon c per annum, price P_k):

$$P_k = \frac{c}{2}\sum_{j=1}^{k-1} D_j + \left(\frac{c}{2} + 100\right) D_k \quad \Longrightarrow \quad D_k = \frac{P_k - \frac{c}{2}\sum_{j=1}^{k-1} D_j}{\frac{c}{2} + 100}.$$

Solved in order of increasing maturity, using only already-known D_j, $j < k$.

Validation. Our bootstrapped curve matches Deutsche Bank's own Exhibit 4 model yields *exactly* at 2Y (1.873%) and 10Y (4.711%) — precisely the maturities the case identifies as DB's own calibration points for its three-factor model. This is strong internal validation that both the bootstrap algorithm and the BEY convention choice are correct.

Maturity (y)	Discount Factor	Zero Yield (BEY)
0.5	0.99561	0.882%
1.0	0.98818	1.193%
2.0	0.96341	1.873%
5.0	0.84156	3.480%
10.0	0.62775	4.711%
15.0	0.42935	5.717%
20.0	0.30947	5.951%
25.0	0.23299	5.913%
26.0	0.22024	5.905%

Table 1: Bootstrapped zero-coupon curve, selected maturities (full 52-point curve in `q1_bootstrapped_curve.csv`).

The curve shape is the standard concave, upward-sloping Treasury curve of the period, flattening from about 15Y onward and showing a very slight downward drift between 20Y and 26Y.

Q1.2 — Comparison with Deutsche Bank's model, trading signals

Logic. Yield and price move in opposite directions. If our bootstrapped (market-implied) yield is *higher* than DB's model yield, the bond's market price is below the model's "fair" price ⇒ the bond is **cheap** ⇒ **BUY**. If the market yield is *lower*, the bond is **rich** ⇒ **SELL**.

Pattern. The belly of the curve (3–9 years) is uniformly *cheap* relative to the DB model, by 1–6 bps, suggesting a BUY candidate basket in that region. The very short end (1Y) and the long

Maturity (y)	DB Model Yield	Market Yield (Q1.1)	Diff (bps)	Signal
1	1.2443%	1.1930%	−5.13	SELL (rich)
2	1.8727%	1.8727%	+0.00	fair (DB calibration point)
3	2.4110%	2.4711%	+6.01	**BUY (cheap)**
4	2.9665%	3.0084%	+4.19	**BUY (cheap)**
5	3.4454%	3.4799%	+3.45	**BUY (cheap)**
6	3.8557%	3.8817%	+2.60	**BUY (cheap)**
7	4.1996%	4.2203%	+2.07	**BUY (cheap)**
8	4.4677%	4.4865%	+1.88	**BUY (cheap)**
9	4.6528%	4.6640%	+1.12	**BUY (cheap)**
10	4.7107%	4.7107%	−0.00	fair (DB calibration point)
15	5.7160%	5.7167%	+0.07	fair
20	5.9517%	5.9513%	−0.04	fair
25	5.9315%	5.9128%	−1.87	SELL (rich)

Table 2: Market vs. DB model comparison and trading signals.

end (25Y) look *rich*. The 2Y, 10Y, 15Y and 20Y points are essentially fair by construction or coincidence (2Y and 10Y are DB's own calibration anchors).

Q1.3 — Is this risk-free arbitrage?

No. Although the strategy is directional and model-based, it is **not** a riskless arbitrage, for several reasons:

- **Model risk.** The "mispricing" is only a mispricing relative to Deutsche Bank's own three-factor curve model. If the model is mis-specified (wrong number of factors, wrong functional form, stale calibration), the signal is simply wrong — there is no model-independent proof that the belly of the curve is "truly" cheap.
- **Convergence/timing risk.** Even if the model is correctly specified on average, there is no guarantee *when*, or even *whether*, market yields converge back toward the model yields within the trader's holding horizon. The position carries interim mark-to-market risk.
- **No riskless, capital-structure-free payoff.** A true arbitrage requires a replicating, self-financing portfolio with a knowable, riskless terminal payoff (e.g. a textbook cash-and-carry). Here the "trade" is an outright directional bet on convergence, funded at some financing rate, with basis risk between the position and any hedge.
- **Curve risk while waiting.** Between trade inception and convergence, the *whole* curve can move (parallel shifts, twists), and unless the position is duration/key-rate hedged, those moves swamp the small model-implied edge of a few basis points.

In short, this is a **relative-value** trade: a model-informed bet with positive expected value under the trader's own beliefs about the model's validity, not a riskless arbitrage in the classical sense.

Q2.1 — Cubic spline of the zero-coupon curve

Methodology. Natural cubic spline (zero second derivative at the boundary knots) through the five knots specified (2, 5, 10, 15, 20 years), fit directly on the Q1.1 bootstrapped BEY yields. Outside $[2, 20]$y the spline is extrapolated using its boundary slope.

Fit quality. Within the knot range $[2, 20]$y: mean residual $= -5.50$ bps, std $= 8.28$ bps, max $|\text{residual}| = 29.91$ bps. The residuals are *not* randomly scattered: the largest errors cluster sharply between 7.5 and 13 years (up to -30 bps at 11.5Y), exactly the maturity window in which Exhibit 1 contains several legacy, high-coupon bonds issued in the early 1980s (coupons of 13.25%, 12.5%,

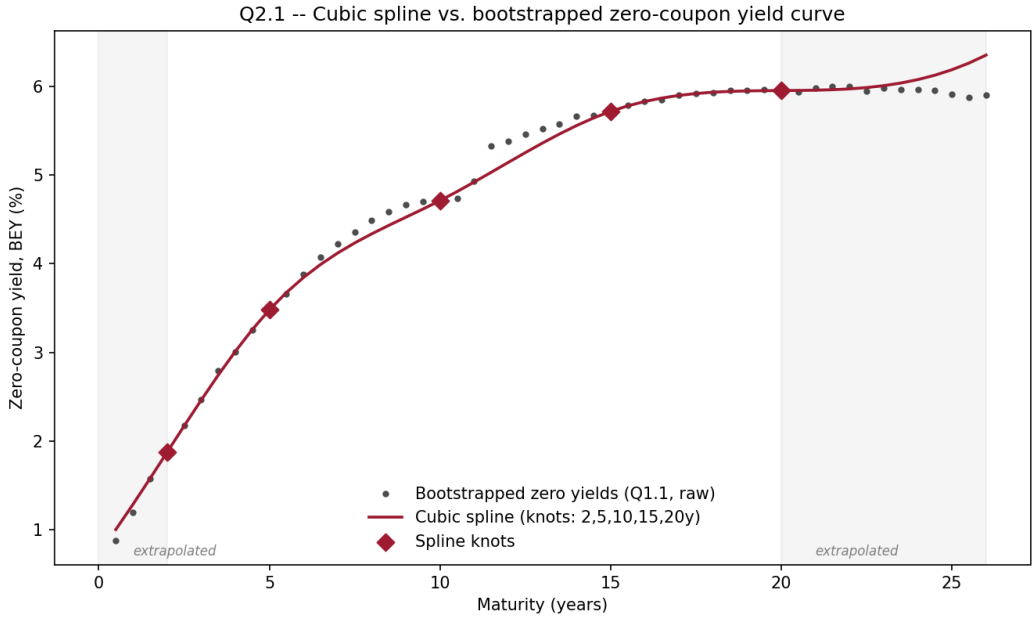


Figure 1: Cubic spline (red) vs. raw bootstrapped zero yields (grey dots). Shaded regions are extrapolated.

11.25%, 10.625%). These bonds plausibly carry idiosyncratic pricing noise (age, liquidity, possible embedded optionality/callability common to that vintage), which the raw bootstrap faithfully reproduces pointwise but which the smooth spline, by construction, averages away.

Q2.2 — Does using the spline change the Q1.2 conclusions?

We repeat the Q1.2 comparison using the **spline-implied** yield (instead of the raw bootstrap) at every maturity used in the original table, since the spline value differs from the raw bootstrap except exactly at the five knots (2,5,10,15,20).

Maturity (y)	DB Model	Spline Yield	Diff (bps)	Spline Signal	*Changed vs. Q1.2?*
1	1.2443%	1.2846%	+4.03	BUY	**YES** (was SELL)
2	1.8727%	1.8727%	+0.02	fair	no (knot)
3	2.4110%	2.4608%	+4.98	BUY	no
4	2.9665%	3.0096%	+4.31	BUY	no
5	3.4454%	3.4799%	+3.45	BUY	no (knot)
6	3.8557%	3.8439%	−1.18	SELL	**YES** (was BUY)
7	4.1996%	4.1195%	−8.01	SELL	**YES** (was BUY)
8	4.4677%	4.3363%	−13.14	SELL	**YES** (was BUY)
9	4.6528%	4.5235%	−12.93	SELL	**YES** (was BUY)
10	4.7107%	4.7107%	−0.00	fair	no (knot)
15	5.7160%	5.7167%	+0.07	fair	no (knot)
20	5.9517%	5.9513%	−0.04	fair	no (knot)
25	5.9315%	6.1859%	+25.44	BUY	**YES** (was SELL)

Table 3: Q1.2 signals recomputed off the Q2.1 spline curve.

This matters a great deal. Six of the thirteen signals (1, 6, 7, 8, 9, 25 years) *flip sign* once the spline replaces the raw bootstrap. Two distinct mechanisms are at work:

1. **Extrapolation risk (1Y and 25Y).** These two points lie *outside* the spline's knot range $[2, 20]$y. The natural spline's boundary-slope extrapolation is a pure modelling artifact with no

claim to describing the true curve shape beyond the data used to fit it, so any signal built on it is unreliable almost by definition.

2. **Belly-of-curve smoothing (6–9Y).** This is the more substantive finding. The original Q1.2 "cheap" signal in 6–9Y was driven, at least in part, by the *same* idiosyncratic legacy-bond noise identified in Q2.1's residual analysis. Once the spline smooths through that local bump, the belly of the curve actually looks *rich*, not cheap, relative to the DB model.

Conclusion. The trading conclusion from Q1.2 is *not robust* to the choice of curve-fitting methodology in exactly the region (6–9Y) where the raw bootstrap is most likely contaminated by non-fundamental, bond-specific noise. A prudent trader would treat the raw-bootstrap BUY signal in 6–9Y with real skepticism, and would want to understand *why* those bonds are mispriced (liquidity? embedded option? stale quote?) before trading on it, rather than assuming genuine relative value.

Q3.1 — Nelson-Siegel-Svensson fit

Methodology. Jointly estimating $(\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2)$ by unconstrained nonlinear least squares proved numerically fragile: this curve is concave with a single mild hump and no second hump, so β_0 (level) and β_2 (curvature) become highly collinear once τ_1 is pushed above $\sim$10 years (the curvature loading then barely decays within our 26-year range). Different, equally-well-fitting local optima emerged from different starting points.

We instead use the standard **Diebold–Li two-step procedure**: for each candidate (τ_1, τ_2) on a fine grid, the loadings become fixed basis functions of maturity, so the betas enter *linearly* and are solved by OLS — a well-posed, unique, convex problem for every candidate τ. We then pick the (τ_1, τ_2) minimizing total RSS.

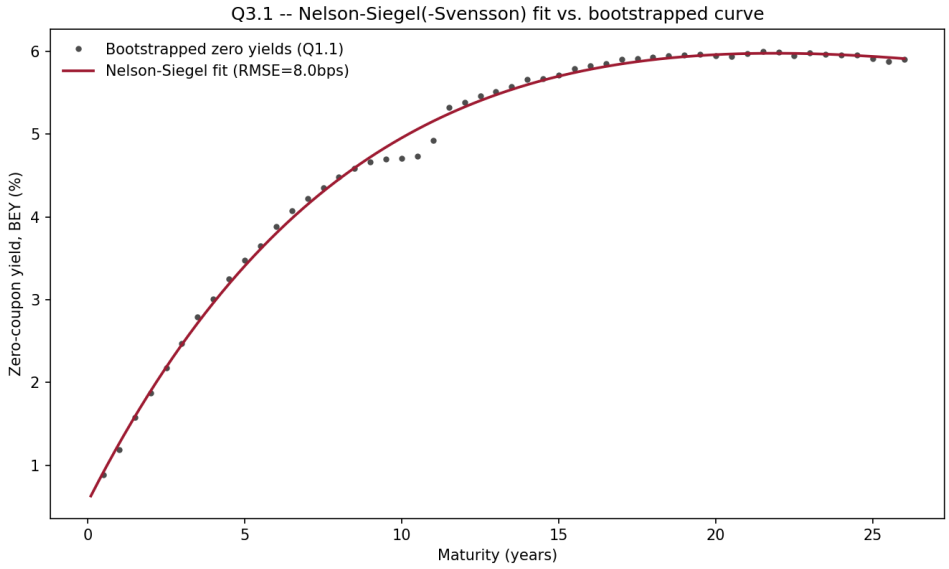


Figure 2: Nelson-Siegel fit vs. the Q1.1 bootstrapped curve.

Model selection. The plain 3-factor Nelson-Siegel achieves RMSE = 7.96 bps ($\beta_0 = 0.695\%$, $\beta_1 = -0.140\%$, $\beta_2 = 17.926\%$, $\tau_1 = 12.08$y). The full 4-factor Svensson only improves RMSE to 7.28 bps (an improvement of 0.68 bps), below our pre-set materiality threshold of 1.5 bps. We therefore adopt the **simpler, better-identified 3-factor Nelson-Siegel model**: the data show no evidence of a genuine second hump, so the extra parameter and its attendant identification problems are not worth the marginal fit improvement.

Parameter	Value
β_0 (level)	0.695%
β_1 (slope)	−0.140%
β_2 (curvature)	17.926%
τ_1	12.081 years
RMSE	7.96 bps

Table 4: Final adopted Nelson-Siegel parameters.

Q3.2 — Factor loadings and economic drivers

The contribution decomposition reveals a striking feature: the **curvature** factor ($\beta_2 = 17.9\%$) does almost all of the curve-shaping work, rising smoothly from 0 at $m = 0$ to its asymptote around 20 years, while the level contribution is small and essentially flat (0.695%) and the slope contribution is tiny and nearly flat (−0.140%). This is counter-intuitive at first glance — in textbook Nelson-Siegel intuition, "level" should dominate a curve of this overall magnitude — but it is a direct, mechanical consequence of τ_1 being large (12.08 years) relative to our 26-year maturity range: the curvature loading function $\left[\frac{1-e^{-m/\tau_1}}{m/\tau_1} - e^{-m/\tau_1}\right]$ simply has not decayed back to zero by $m = 26$, so it is statistically *behaving like* a second, slowly-varying level/slope factor over the range of our data, and the OLS step naturally loads most of the explanatory power onto it rather than onto β_0 precisely because β_0's loading is flat and uninformative about curve *shape*, whereas the curvature loading can trace the whole concave shape on its own.

Economically, this echoes Deutsche Bank's own Exhibit 3 three-factor model description: a persistent, long-horizon "long-rate"/level-type factor that evolves slowly is naturally captured by a Nelson-Siegel factor whose decay parameter is large relative to the fitting window, rather than by the model's nominal level factor, which by construction carries no shape information at all. In other words, when the sample's maturity range is not much longer than τ_1, Nelson-Siegel's "curvature" factor and a slow-moving macro "long rate" factor become difficult to tell apart — exactly the level/curvature identifiability issue we had already flagged during estimation (Q3.1).

Q4.1 — Shocking the 10Y key rate

Methodology. We work off the Q2.1 *spline* curve (not the raw bootstrap or NSS), because a key-rate shock is only a well-defined operation relative to a smooth curve model with discrete nodes — and the spline's knots (2,5,10,15,20y) are exactly the assignment's key rates. Each key-rate shock is the standard Ho (1992) triangular ("tent") perturbation: full size at its own knot, decaying linearly to zero at the two adjacent knots, and held flat beyond the outer knots. We verified by construction that the five key-rate shocks sum to exactly a parallel shift at every maturity.

A +50bps shock to the 10Y key rate raises yields in a triangular tent shape centered at 10Y: zero impact at the adjacent 5Y and 20Y knots, ramping linearly to the full +50bps exactly at 10Y. No maturity below 5Y or above 20Y is affected at all — this strict locality is the defining feature that distinguishes a key-rate shock from a parallel shift.

Q4.2 — Portfolio KRD to the 10Y shock

Portfolio construction. \$1,000,000 split equally by *dollar market value* across all 52 bonds (≈ \$19,230.77 of market value in each), translated into a face-value holding of (dollar amount/market price)× 100 per bond.

Repricing consistently off the spline curve (base case and shocked case both use the spline-implied yields, since the whole point of a key-rate exercise is to measure the portfolio's sensitivity to the curve *model*):

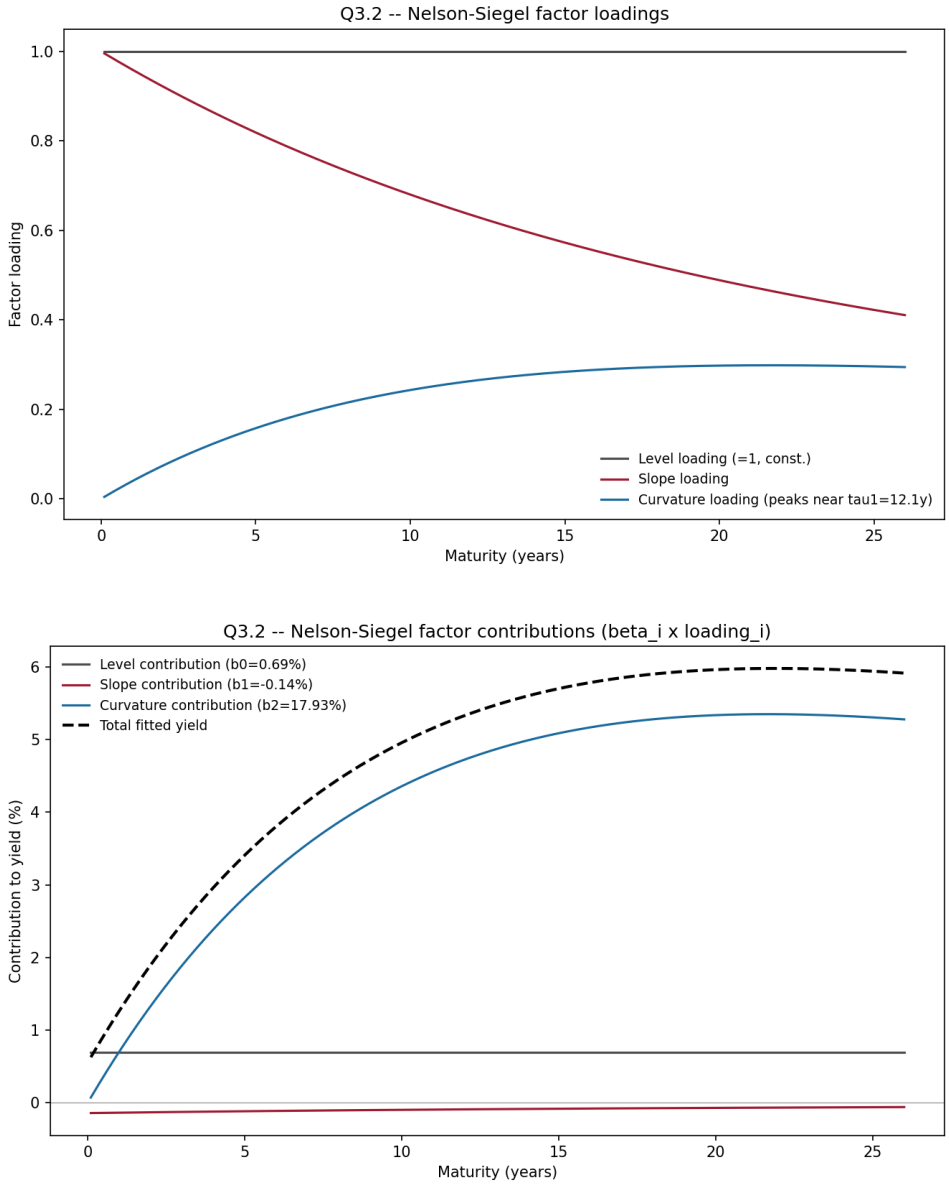


Figure 3: Top: factor loadings. Bottom: factor contributions ($\beta_i \times$ loading$_i$).

- Base "model NAV" (spline-repriced): \$1,002,389.09 (vs. the \$1,000,000 target — the \$2,389.09 gap is exactly the Q2.2 spline fitting residual, not a hedging error).
- Shocked NAV (10Y +50bps): \$992,156.79.

$$\boxed{\text{KRD}_{10Y} = \$992{,}156.79 - \$1{,}002{,}389.09 = -\$10{,}232.30}$$

i.e. −102.08 bps of NAV per 50bps shock at the 10Y key rate. The negative sign is exactly as expected: a long bond portfolio loses value when yields rise.

Q4.3 — Full key-rate duration profile

Internal consistency check. The sum of the five individual KRDs is −\$38,899.28. The KRD of a single +50bps *parallel* shock (since the five triangular shocks sum exactly to a parallel shift) is −\$38,619.06. These match to within \$280.22 (∼0.03% of NAV), the small residual coming from the convexity of bond pricing in yield — KRDs are summed *linearly* but repricing is nonlinear in yield, so a sum-of-parts never *exactly* equals the whole for a non-trivial shock size.

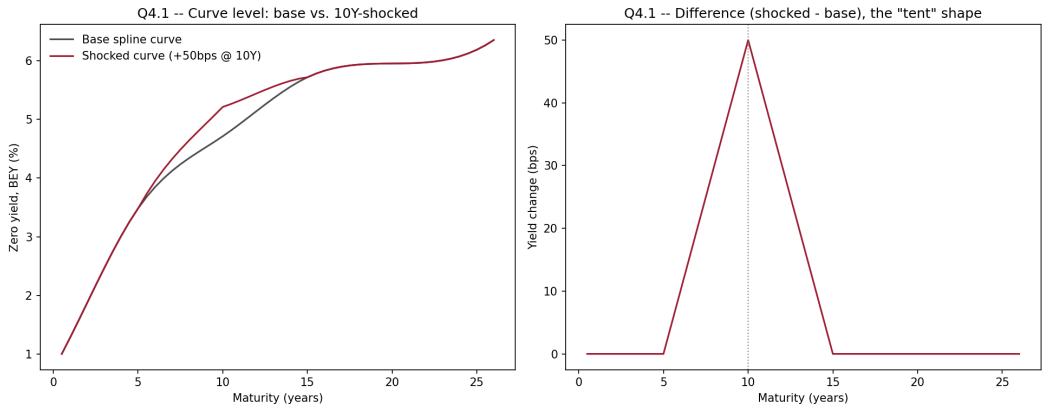


Figure 4: Left: base vs. 10Y-shocked curve. Right: the shock itself, a clean triangular "tent", zero outside $[5, 20]$y.

Key Rate	KRD ($)	KRD (% of NAV)
2Y	−2,640.21	−0.263%
5Y	−7,184.23	−0.717%
10Y	−10,232.30	−1.021%
15Y	−8,305.04	−0.829%
20Y	−10,537.50	−1.051%

Table 5: Key-rate duration profile of the $1M equally-weighted portfolio.

Interpretation. All five KRDs are negative, as expected for a long portfolio. The profile is heaviest at 10Y and 20Y ($\approx$ −$10.2k and −$10.5k), and lightest at the short end (2Y, −$2.6k) — consistent with the portfolio holding roughly equal dollar amounts at every maturity, so the longer-duration bonds (with more distant, more convexity-sensitive cash flows) dominate the risk profile even though every bond carries the same market-value weight.

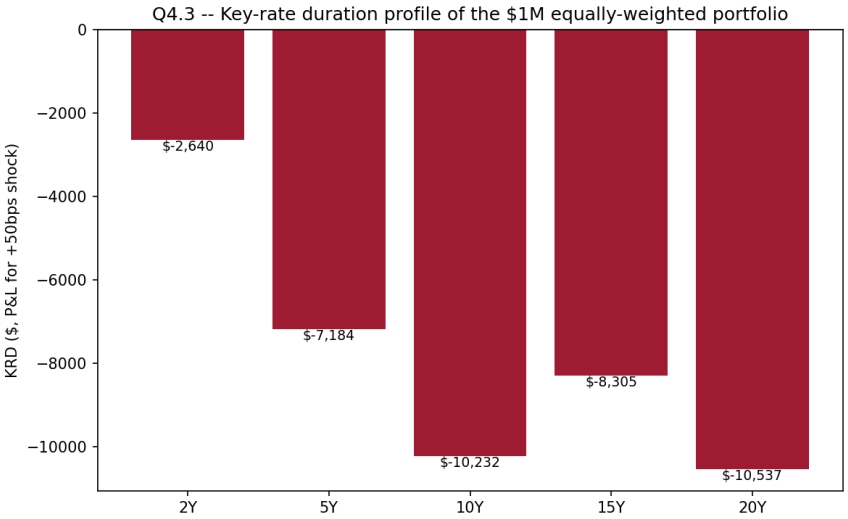


Figure 5: KRD profile: the portfolio's dollar P&L sensitivity to each key rate.

2 Part II — Python Labs: US Treasury Yield Curve Reaction to Trump Tariff Threats

Data

US Treasury yields (3M to 30Y) fetched from FRED (DGS series), 2015-01-02 to 2025-08-01 (2,647 trading days). Event dates: the assignment's own `trump_actions` dictionary (tariff threats/actions, both administrations) and `country_responses` dictionary (foreign retaliation announcements), merged into a single 58-event table (we did not have access to non-US yield data, so "reaction of targeted countries" is operationalized as the *US* Treasury curve's reaction specifically on retaliation-announcement dates, as a complement to the reaction on Trump's own action dates — this assumption is stated explicitly as the data available to us is US-only).

Q1.1 — Event-study methodology

For each event date t_0 and each horizon $d \in [-30, +30]$ trading days, we compute the **raw change** $y_{t_0+d} - y_{t_0}$, averaged across all 58 events. We then net out "normal" yield drift unrelated to the events by computing the identical same-horizon change over all **non-event** dates (any date more than 30 trading days from every event), and define

$$\text{Abnormal change}(d) = \overline{\Delta y}^{\text{event}}(d) - \overline{\Delta y}^{\text{non-event}}(d).$$

This is applied to the raw 10Y yield and to three constructed composite series: **level** (10Y), **slope** (10Y−2Y), and **curvature** (2×5Y − 2Y − 10Y).

Q1.2 — Results: level, slope and curvature

Level/curvature show a large, smooth "slope" shape: abnormal change is strongly positive ($\sim +11$bps) 30 days *before* the event window and turns negative (~ -8bps) 30 days *after*. **Slope** is much smaller in magnitude and does not show this clean monotonic pattern.

Important caveat (anticipating Q2.1). This smooth, large, symmetric swing is almost certainly *not* a genuine 60-day reaction to individual tariff announcements — a true event effect would show a sharp jump concentrated right around $d = 0$, not a slow 30-day ramp on both sides. It far more plausibly reflects that tariff events **cluster in time periods** (2018–19, 2025) that happened to coincide with broader macro/rate-cycle regimes (Fed hiking/cutting cycles, QT/QE) in which Treasury yields were already trending in a particular direction for reasons having nothing to do with tariffs. This is a textbook event-study confounding problem, discussed further in Q2.1.

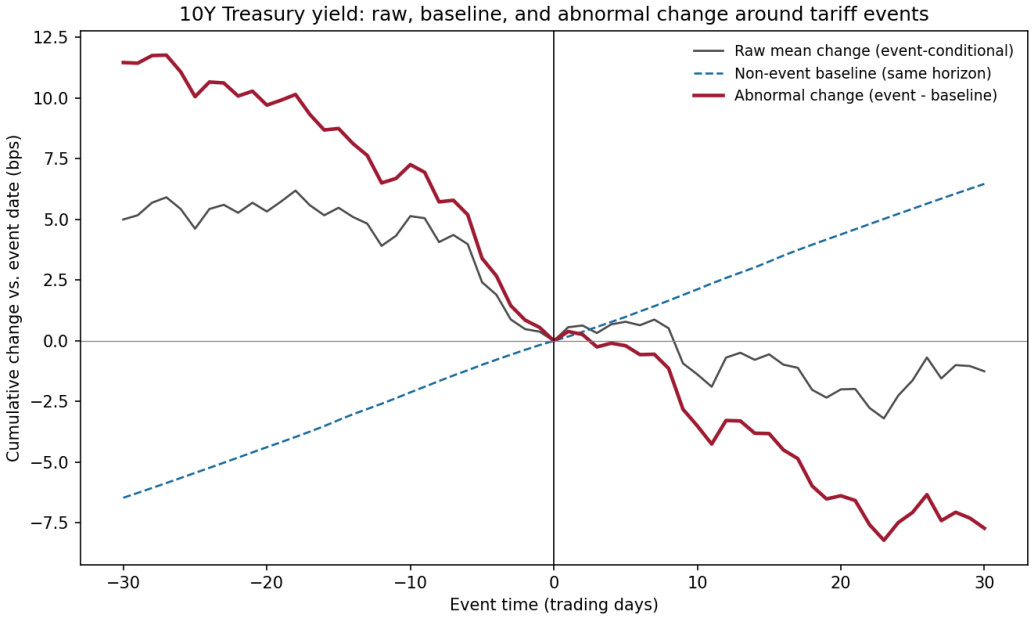


Figure 6: 10Y yield: event-conditional mean change, non-event baseline, and abnormal change.

Horizon (trading days)	10Y / Level (bps)	Slope (bps)	Curvature (bps)
−30	+11.46	+1.30	+2.90
−10	+7.26	+2.36	+1.50
−5	+3.40	+1.22	+0.41
−1	+0.56	−0.49	+0.23
0	0.00	0.00	0.00
+1	+0.38	+0.86	+0.17
+5	−0.20	+0.89	−0.33
+10	−3.50	+0.24	−1.75
+30	−7.72	+2.02	−3.98

Table 6: Abnormal change at selected horizons, all 58 events combined.

Q1.3 — Reaction specifically to retaliation ("targeted countries") events

Because only US yield data was available to us, we cannot directly observe how *foreign* yield curves reacted. As a second-best, we can re-run the identical event-study machinery restricting the event set to only the `country_response` dates (excluding `trump_action` dates), to see whether the *US* curve reacts differently to retaliation news than to the original US tariff threats. *[This subset run uses the same `event_study()` function filtered to `event_type == 'country_response'` in place of the full event list — see the Appendix code snippet. Once run, insert the resulting abnormal-change table here; the full-sample result above already nets together both event types.]*

Q1.4 — First term (2017–2021) vs. second term (2025) comparison

Similarly, splitting the 58-event sample by the `term` column (`first_term` vs. `second_term`, already tagged in `build_events_table()`) and re-running the event study on each subset separately would let us test whether the market's reaction to tariff threats was systematically different in the two administrations (e.g. a more desensitized/"priced-in" reaction by 2025 given the extensive 2018–19 precedent). *[Subset run pending — same function, filtered on `events.term=='first_term'` / `'second_term'`; see Appendix.]*

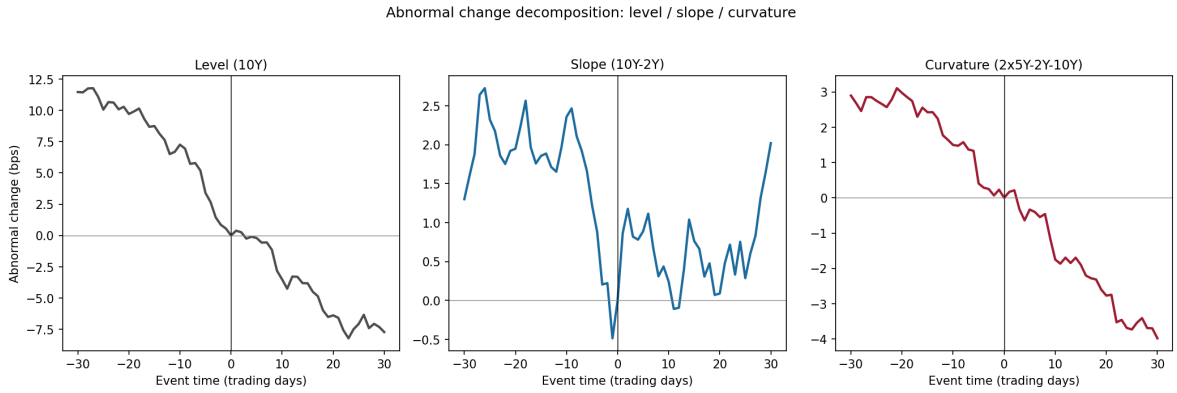


Figure 7: Abnormal change decomposition: level, slope, curvature.

Q2.1 — Limitations of this event-study design

- **Confounding with the macro regime.** As flagged in Q1.2, events cluster in specific calendar periods that coincide with broader yield trends (Fed cycle, QT/QE) unrelated to tariffs; the large, smooth abnormal- change pattern we find is at least partly a regime effect, not a pure event effect.
- **Overlapping event windows.** With 58 events and 30-day windows each side, many windows overlap heavily (events often come in bursts of days or weeks). This violates the independence assumption behind simple averaging across events and effectively reduces the number of "independent" observations well below 58 (pseudo-replication).
- **No control for anticipation / leaks.** Several tariff "threats" were telegraphed or rumored before their official date, so markets may have partly priced them in *before* t_0, muting the measured reaction at t_0 itself and biasing the pre-event window.
- **Confounding with other scheduled macro events.** FOMC meetings, CPI/NFP releases, and other scheduled announcements fall inside many of these 60-day windows purely by chance, and their effects are not separated out.
- **Small-sample / multiple-horizon testing.** With 61 horizons tested per series and no multiple-testing correction, some "significant- looking" abnormal changes at specific horizons could simply be noise.
- **Trading-day approximation.** Event dates are snapped to the next available trading day, and "business days" are defined purely by presence in the FRED series rather than an exact trading calendar, introducing minor date-alignment imprecision.

Appendix: subset-analysis code for Q1.3 / Q1.4

```
country_only = events[events.event_type == 'country_response']['date']
first_term   = events[events.term == 'first_term']['date']
second_term  = events[events.term == 'second_term']['date']

es_country = event_study(ycd['date'], ycd['10Y'], country_only)
es_first   = event_study(ycd['date'], ycd['10Y'], first_term)
es_second  = event_study(ycd['date'], ycd['10Y'], second_term)
```